

Planner View: Distributed Order Management Recommendation

Project Overview

This project presents a Hybrid Quantum–Classical Optimization Framework developed for the WISER × Nestlé Quantum Challenge 2026. The objective is to improve customer order fulfillment by identifying the most suitable distribution center (DC) for each order while considering operational constraints and business priorities. Unlike traditional rule-based assignment methods, the proposed framework evaluates multiple feasible distribution centers simultaneously and recommends assignments that balance revenue, shipping cost, inventory availability, throughput capacity, dock capacity, and fulfillment penalties.

Business Problem

In a distributed supply chain, the default distribution center may not always be capable of fulfilling an order due to inventory shortages or operational capacity limitations. Manual reassignment decisions become increasingly difficult as the number of customer orders and distribution centers grows.

The proposed framework automatically recommends the most suitable distribution center while ensuring that:

- Each order is assigned to at most one distribution center.
- Inventory and operational capacity constraints are respected.
- Shipping costs and fulfillment penalties are minimized.
- Business value is maximized.

Solution Summary

The optimization workflow consists of five major stages:

1. Data preprocessing and validation
2. Feasible candidate generation
3. Adaptive candidate reduction
4. Hybrid quantum–classical optimization
5. Business repair and constraint validation

A business repair layer ensures that all final recommendations satisfy operational constraints before they are presented to planners.

Key Results

Metric	Result
Feasible Candidate Assignments	133,495
Candidate Assignments after Reduction	65,857
Candidate Reduction	50.67%
Raw Quantum Assignments	15
Final Business Assignments	12
Constraint Violations After Repair	0

The proposed framework provides several advantages for supply chain planners:

- Faster evaluation of multiple distribution center options.
- Reduced manual decision-making effort.
- Improved utilization of inventory and operational capacity.
- Lower shipping costs and fulfillment penalties.
- Guaranteed business-feasible order assignments through post-processing validation.
- Scalable workflow that can support future quantum computing technologies.

Recommendation

The proposed Hybrid Quantum–Classical Optimization Framework demonstrates that advanced optimization techniques can improve Distributed Order Management by combining efficient classical preprocessing with quantum optimization and business validation.

While current quantum hardware limits large-scale execution, the framework establishes a practical workflow that can be expanded as quantum computing technology matures. The approach provides reliable, explainable, and business-feasible recommendations for customer order reassignment.